

Section 3.2

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$$1. f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \frac{(4 - (x+h)^2) - (4 - x^2)}{h} = \lim_{h \rightarrow 0} \frac{4 - x^2 - 2xh - h^2 - 4 + x^2}{h} = \lim_{h \rightarrow 0} \frac{-2xh - h^2}{h} = \lim_{h \rightarrow 0} -2x - h = -2x$$

$$\therefore f'(-3) = -2(-3) = 6$$

$$f'(0) = -2(0) = 0$$

$$f'(1) = -2(1) = -2$$

$$20. \frac{dy}{dx} = \lim_{h \rightarrow 0} \frac{(1 - \frac{1}{x+h}) - (1 - \frac{1}{x})}{h} = \lim_{h \rightarrow 0} \frac{\frac{1}{x} - \frac{1}{x+h}}{h} = \lim_{h \rightarrow 0} \frac{1}{x(x+h)} = \frac{1}{x^2}$$

$$\therefore \left. \frac{dy}{dx} \right|_{x=\sqrt{3}} = \frac{1}{(\sqrt{3})^2} = \frac{1}{3}$$

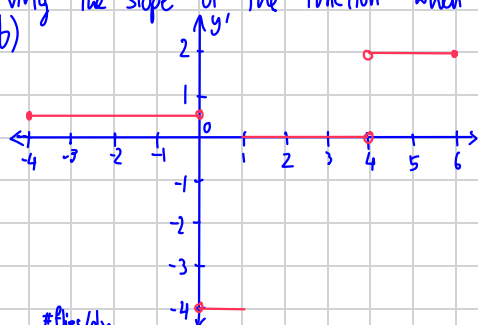
27. b

28. a

29. d

30. c

31. a) f' is not defined at $(0, 2)$, $(1, -2)$, $(4, -2)$. At point $(0, 2)$, we get that $f'(0^-) \neq f'(0^+)$, also, at point $(1, -2)$, $f'(1^-) \neq f'(1^+)$ and at point $(4, -2)$, $f'(4^-) \neq f'(4^+)$. It could be seen by observing the slope of the function when approaching $x=0, 1, 4$ from each side.



b) Fastest: during day 25
Slowest: during day 0, day 50

43. Differentiable at every point in D

44. Differentiable at every point in D

46. Differentiable at all points except $x=-1, 0, 2$
Continuous but not differentiable at $x=-1$

Neither continuous nor differentiable at $x=0, 2$

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47. Differentiable at all x except $x=0$

Continuous but not differentiable at $x=0$

48. Differentiable at all x except $x=-2, 2$

Continuous but not differentiable at $x=-2, 2$

57. $\lim_{t \rightarrow 0} \frac{g(t)}{h(t)}$ can exist, and can be other values than zero. It did not state that $g(t)$ and $h(t)$ are continuous or differentiable, so we can assume $g(t)$ and $h(t)$ to be any function and leave a hole at $t=0$ for both functions so that $g(0)=h(0)=0$.

58. a) As $|f(0)| \leq 0^2$ or $0 \leq f(0) \leq 0$, we get that $f(0)=0$.

$$\therefore f'(0) = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{f(h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{f(h)}{h}$$

For $|h| \leq 1$, we get $-h^2 \leq f(h) \leq h^2$, so $-|h| \leq \frac{f(h)}{h} \leq |h|$ and by sandwich theorem, we get $\lim_{h \rightarrow 0} \frac{f(h)}{h} = 0$, or $f'(0)=0$.

b) For $x \neq 0$, $|f(x)| = |x^2 \sin(\frac{1}{x})| = |x^2| |\sin(\frac{1}{x})| \leq |x^2| \cdot 1 = x^2$
and since $|f(0)| = 0 \leq 0^2$, it follows 58a).
Thus, $f(x)$ is differentiable at $x=0$ and $f'(0)=0$

Section 3.3

6. $y' = \frac{1}{3}(3x^2) + \frac{1}{2}(2x) + \frac{1}{4}(1) = x^2 + x + \frac{1}{4}$, $y'' = 2x + \frac{1}{2}$

11. $r' = \frac{1}{3}(\frac{-2}{s^3}) - \frac{5}{2}(\frac{-1}{s^3}) = \frac{-2}{3s^3} + \frac{5}{2s^3}$, $r'' =$

17. $y' = \frac{3(2x+5) - 2(3x-2)}{(3x-2)^2} = \frac{19}{9x^2 - 12x + 4}$

20. $f'(t) = \frac{(2t)(t^2+t-2) - (2t+1)(t^2-1)}{(t^2+t-2)^2}$

21. $v' = \frac{(1+t^2)(-1) - (1-t)(2t)}{(1+t^2)^2}$

29. $y' = 9x^3 - 3x - 1$, $y'' = 6x^2 - 3$, $y^{(3)} = 12x$, $y^{(4)} = 12$, $y^{(n)} = 0$ for all $n \geq 5$

37. $w = (1 + \frac{1}{3z})(3-z) = \frac{8}{3} - z + \frac{1}{z}$

$w' = -1 - \frac{1}{z^2}$, $w'' = \frac{2}{z^3}$

38. $w = (z+1)(z-1)(z^2+1) = z^4 - 1 \rightarrow w' = 4z^3$, $w'' = 12z^2$

41. a) $\frac{d}{dx}(uv)|_{x=0} = u \frac{dv}{dx}|_{x=0} + v \frac{du}{dx}|_{x=0} = u(0)v'(0) + v(0)u'(0) = 5 \cdot 2 + (-1)(-3) = 13$

b) $\frac{d}{dx} \left(\frac{u}{v} \right) \Big|_{x=0} = \frac{v \frac{du}{dx} \Big|_{x=0} - u \frac{dv}{dx} \Big|_{x=0}}{v^2 \Big|_{x=0}} = \frac{v(0)u'(0) - u(0)v'(0)}{(v(0))^2} = \frac{(-1)(-3) - 5 \cdot 2}{(-1)^2} = -7$

c) $\frac{d}{dx} \left(\frac{v}{u} \right) \Big|_{x=0} = \frac{u \frac{dv}{dx} \Big|_{x=0} - v \frac{du}{dx} \Big|_{x=0}}{u^2 \Big|_{x=0}} = \frac{u(0)v'(0) - v(0)u'(0)}{(u(0))^2} = \frac{5 \cdot 2 - (-1)(-3)}{5^2} = \frac{7}{25}$

41. d) $\frac{d}{dx}(7v-2u)|_{x=0} = \frac{d}{dx}(7v)|_{x=0} - \frac{d}{dx}(2u)|_{x=0} = 7\frac{dv}{dx}|_{x=0} - 2\frac{du}{dx}|_{x=0} = 7v'(0) - 2u'(0) = 7 \cdot 2 - 2 \cdot (-3) = 20$

43. Slope of tangent line of the graph $y = x^3 - 4x + 1$ at $(2, 1)$ equals
 $y'(2) = \frac{dy}{dx}|_{x=2} = 3x^2 - 4|_{x=2} = 3(2)^2 - 4 = 8$

Hence, slope of the line perpendicular to this tangent line $= -\frac{1}{8}$

And the equation of this perpendicular line is $y - 1 = -\frac{1}{8}(x - 2)$
 $y = -\frac{1}{8}x + \frac{5}{4}$

48. The slope of both tangent lines are equal, hence

$$\frac{d}{dx}(x^2 + 2x + b)|_{x=1} = \frac{d}{dx}(cx - x^2)|_{x=1}$$

$$2(1) + 2 = c - 2(1)$$

$$2 + 2 = c - 2$$

$$a = c - 4$$

Since $y = cx - x^2$ passes $(1, 0)$, it means $c(1) - 1^2 = 0$ or $c = 1$

So, $a = 1 - 4 = -3$ since $y = x^2 + 2x + b$ also passes $(1, 0)$, we get that

$$0 = 1^2 + (-3)(1) + b \rightarrow b = 2$$

Hence, $a = -3$, $b = 2$, $c = 1$ #

55. $\lim_{x \rightarrow 1} \frac{x^{50} - 1}{x - 1} = \lim_{x \rightarrow 1} \frac{\frac{d}{dx}(x^{50} - 1)}{\frac{d}{dx}(x - 1)} = \lim_{x \rightarrow 1} \frac{50x^{49}}{1} = 50$

56. $\lim_{x \rightarrow -1} \frac{x^{2/9} - 1}{x + 1} = \lim_{x \rightarrow -1} \frac{\frac{d}{dx}(x^{2/9} - 1)}{\frac{d}{dx}(x + 1)} = \lim_{x \rightarrow -1} \frac{\frac{2}{9}x^{-7/9}}{1} = \frac{2}{9}(-1) = -\frac{2}{9}$

57. We have to make sure that $g'(0^-) = g'(0^+)$ so that $g(x)$ is differentiable for all x -values

$$\therefore g'(0^-) = g'(0^+)$$

$$\frac{d}{dx}(2x)|_{x=0} = \frac{d}{dx}(x^2 - 3x)|_{x=0}$$

$$2 = 2x - 3|_{x=0} = -3$$

$$\therefore a = -3$$

58. To make $f(x)$ differentiable for all x , we must get that $f'(-1^-) = f'(-1^+)$

$$\therefore f'(-1^-) = f'(-1^+)$$

$$\frac{d}{dx}(bx^2 - 3)|_{x=-1} = \frac{d}{dx}(2x + b)|_{x=-1}$$

$$2bx|_{x=-1} = 2$$

$$-2b = 2 \rightarrow a \text{ must be equal to } -2b \text{ to make } f(x) \text{ differentiable}$$

Section 3.4

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3. a) $S(0) = 0 \text{ m}$, $S(3) = -27 + 27 - 9 = -9 \text{ m} \therefore V_{\text{average}} = \frac{S(3) - S(0)}{3 - 0} = \frac{-9}{3} = -3 \text{ m/s}$
 b) $V(t) = \frac{dS(t)}{dt} = -3t^2 + 6t - 3 \rightarrow V(0) = -3 \text{ m/s}$, $V(3) = -3(3)^2 + 6(3) - 3 = -12 \text{ m/s}$
 $a(t) = \frac{dV(t)}{dt} = -6t + 6 \rightarrow a(0) = 6 \text{ m/s}^2$, $a(3) = -6(3) + 6 = -12 \text{ m/s}^2$
 c) The body changes direction when $V(t) = 0$. Hence, $-3t^2 + 6t - 3 = 0$
 $-3(t-1)^2 = 0 \rightarrow t = 1$

$\therefore$ The body changes the direction when $t = 1 \text{ s}$

7. a) $V(t) = \frac{dS(t)}{dt} = 3t^2 - 12t + 9$

When $V(t) = 0 \rightarrow 3t^2 - 12t + 9 = 0$

$3(t-3)(t-1) = 0 \rightarrow t = 1, 3$

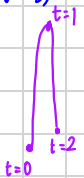
$a(t) = \frac{dV(t)}{dt} = 6t - 12 \rightarrow a(1) = 6(1) - 12 = -6$, $a(3) = 6(3) - 12 = 6$

$\therefore$ When the velocity is zero, the acceleration will be -6 (at $t = 1$) and 6 (at $t = 3$)

b) $a(t) = 0 \rightarrow 6t - 12 = 0 \rightarrow t = 2 \therefore S(2) = 9^3 - 6(2^2) + 9(2) = 2 \text{ m}$

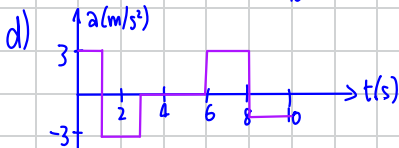
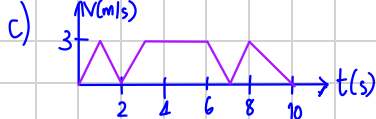
c) As $V(1) = 0$, we can see that the body changes its direction at $t = 1$.

$\therefore$ Total distance traveled by the body from $t = 0$ to $t = 2 = S(0 \rightarrow 1) + S(1 \rightarrow 2)$
 $= S(1) - S(0) + S(1) - S(2)$
 $= 4 - 0 + 4 - 2$
 $= 6 \text{ m}$



15. a) $t = 2$ and $t = 7$ ($v = 0$)

b) From $t = 3$ to $t = 6$



21. A : Velocity, B : Acceleration, C : Distance

$\rightarrow$ B must be acceleration since if B is velocity or distance, its derivative will be zero, but there are no horizontal lines on the figure. Then we get A to be velocity since the derivative of quadratic terms will be linear terms ($\frac{d}{dx}(2x^2 + bx + c) = 2ax + b$), and since B is linear, we get that the graph that represents quadratic equation, or parabola-like, which is A, will be the graph of velocity. That remains C as the graph of distance, since y values of the graph C increases when velocity is greater than 0, decreases when the velocity is less than 0, and the graph C has a turning point when velocity passes 0.

Section 3.4 (Continue)

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23. a) Average cost per machine = $\frac{C(100) - C(0)}{100} = \frac{2000 + 10000 - 1000 - 2000}{100} = \90
 b) Marginal cost = $\frac{dC(x)}{dx} = 100 - 0.2x$
 At $X=100$, marginal cost = $100 - 0.2(100) = 80$.
 c) Cost of producing one more washing machine after the first 100 have been made
 = $C(101) - C(100)$
 = $(2000 + 100 \cdot 101 - 0.1 \cdot 101^2) - (2000 + 100 \cdot 100 - 0.1 \cdot 100^2)$
 = $11079.9 - 11000 = \$79.9$
 $\therefore$ The marginal cost when 100 washing machines are produced is nearly equal to the cost of producing one more washing machine.

25. $\frac{db}{dt} = 10^4 - 2 \times 10^3 t = 10000 - 2000t$
 a) $\left. \frac{db}{dt} \right|_{t=0} = 10000 - 2000 \cdot 0 = 10000$ bacterium/hour
 b) $\left. \frac{db}{dt} \right|_{t=5} = 10000 - 2000 \cdot 5 = 0$ bacterium/hour
 c) $\left. \frac{db}{dt} \right|_{t=10} = 10000 - 2000 \cdot 10 = -10000$ bacterium/hour

26. At $h=180$ cm $\rightarrow S = \frac{\sqrt{180}}{60} \sqrt{W} = \frac{\sqrt{5}}{10} \cdot \sqrt{W}$
 $\frac{dS}{dW} = \frac{\sqrt{5}}{10} \cdot \frac{1}{2\sqrt{W}} \rightarrow$ At lower body weights, S increase more rapidly since S is conversely proportional to W .